

Hill and depression

X16 (2016) — English translation

Bodies A and B move along adjacent paths, the length of which ascent L is equal to the length of descent. The height h of the slide and the depth h of the depression are such that $h \ll L$. The initial velocities of v_0 bodies are identical and satisfy the condition: $v_0^2 \gg gh$.

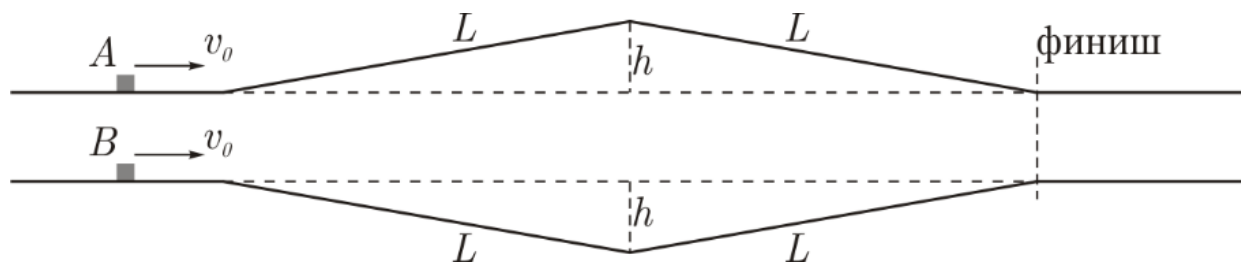


Fig. 1.

A1 Which body will reach the finish line first? Find the delay time Δt of the “slow” body at the finish line. **3.00**

Source: <https://pho.rs/p/4000>